

Priyadharshan L

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Summary

VLSI engineer with a **Tiny Tapeout silicon credit** and hands-on experience in designing and verifying **small-scale IP blocks** — including **RTL design, constrained-random SystemVerilog verification, and RTL-to-GDSII implementation** — using **OpenLane (Sky130)** and **Cadence Genus/Innovus** flows.

Education

College of Engineering, Guindy

Aug 2023-May 2027

B.E. in Electronics and Communication Engineering

- **CGPA:** 7.68 (till 4th August 2026)

Core Competencies

Languages	C
HDLs & Design	Verilog, SystemVerilog, Digital Design, RTL Design
ASIC & Timing	OpenLane (Sky130), Static Timing Analysis, Timing Closure
Architecture & Concepts	RISC-V Architecture, Computer Architecture, CDC, FIFO Design
Tools	Vivado, OpenLane(Sky130), Cadence Genus, Cadence Innovus.

Projects and Experience

RISC-V Core Development – Application-Specific SoC

Ongoing

Team Project — RISC-V Architecture, Design and Verification, SystemVerilog

- Designing and verifying a RV32IM RISC-V core in SystemVerilog; single-cycle implementation being functionally verified currently, with 5-stage pipelined architecture planned next.
- Building core-level understanding of datapath design, control logic, and pipelining trade-offs.
- Planning integration of a lightweight cryptographic IP block into the SoC in a future development phase.

RISC-V based 32-bit ALU unit [Link](#) 🌐

2025

RTL Design(Verilog) ,Verification(SystemVerilog)

- Designed 32-bit RV32I ALU with ImmGen module supporting arithmetic, logic, shift and comparison operations.
- Verified using SystemVerilog constrained-random testbench with assertions and scoreboard.
- Validated corner cases, zero-flag behavior, and register/immediate operation handling.

Asynchronous FIFO - Design and ASIC Implementation (Tiny Tapeout) [Link](#) 🌐

2025

Team Lead — Fifo logic, Verilog, RTL Design, ASIC flow(Openlane), FPGA verification

- Designed parameterized async FIFO using Gray-coded pointers and 2-FF synchronizers for safe CDC.
- Verified full/empty and concurrent operations in Vivado and implemented the design in Basys3 FPGA.
- Taped out via OpenLane (Sky130); closed timing with 0 WNS at a 2.26 ns critical path, 43k μm^2 core area.

Radix-4 Booth Multiplier using CLA (OpenLane and Cadence flows) [Link](#) 🌐

2026

RTL Design, Verilog, Vivado, OpenLane, Sky130 PDK, Cadence

- Designed an 8-bit signed Radix-4 Booth multiplier with a CLA-based final adder in Verilog; implemented using OpenLane (Sky130) and Cadence Genus/Innovus to compare timing performance.
- **OpenLane** : achieved timing closure with 0 WNS, 4.56 ns critical path (10 ns constraint), 0 DRC violations.
- **Cadence** : achieved a 1.278 ns critical path (10 ns constraint) with +8.722 ns setup slack, 0 DRC violations.

RCA vs CLA Adder - ASIC Implementation and analysis [Link](#) 🌐

2026

RTL Design, Verilog, OpenLane, Sky130 PDK, Static Timing Analysis

- Implemented 4-bit and 16-bit RCA and CLA in Verilog; RCA was faster at 4-bit (0.94 ns vs 1.11 ns) while CLA performed better at 16-bit (2.05 ns vs 3.3 ns).
- Reduced 16-bit RCA delay by 27% (3.3 ns $\rightarrow$ 2.41 ns) and showed lower area/power than CLA (96 vs 114 cells, 500 μW vs 520 μW).

Organization and Roles

ECE Association 2024 – 2025

Joint Secretary (Documentation and Permissions)

Robotics Club of CEG 2025

Workshop Organizer